MILITARY SPECIFICATION

SANDER, DISK AND SPINDLE, WOODWORKING

"This specification is approved for use by all Departments and Agencies of the Department of Defense."

1. Scope.

1.1 Scope. This specification covers one type and size combination disk and spindle sander with a 30 inch diameter disk and a single spindle to support sanding drums 9 inches in length, and 2 through 4 inchs in diameter.

2. APPLICABLE DOCUMENTS

2.1. Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services

MILITARY

MIL-STD-1188 - Industrial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and the publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426 appearing at the end of this document or by letter.

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2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402)

(Copies of specifications, standards, handbooks, drawings, and publications required by the manufacturer in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.10 - Machine Tapers, Self-holding and Steep Taper Series

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, New York 10018.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA) STANDARDS

MG-1 - Motors and Generators

IC-1 - Standards for Industrial Controls

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 155 East 44th Street, New York, New York 10017.)

(Industry association specifications and standards are generally available for reference from libraries. They are distributed among technical groups and using Federal Agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 4.3 and 6.2.1), the contractor shall furnish one sample unit for first article inspection and approval.

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3.2 Design. Each sander shall be a self-contained unit having a vertically mounted rotating disk and single oscillating spindle. The design shall include a floor mounting frame, with the disk and disk table mounted on one end and the spindle table mounted on the opposite end. The disk and spindle shall be motor driven. The design shall provide for tilting the disk table and the spindle table and for transverse adjustment of the disk table. All parts subject to wear shall be accessible for repair and replacement.

3.2.1 Reclaimed materials. The machine manufacturer is encouraged to use reclaimed materials to the maximum extent possible without jeopardizing the chemical composition and physical properties of the materials originally selected for the machine construction.

3.2.2 Controls. All operating controls shall be located convenient to the operator's normal work station.

3.2.3 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.3).

3.2.4 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Each recirculating system shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.5 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner that prevents change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

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3.3.3 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.5 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.6 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.3.7 Gears. All gears shall be constructed of a material suitable for their application. Gears shall be applied in a manner to preclude either premature damage, wear or failure.

3.4 Components.

3.4.1 Frame. The frame shall support all of the machine components specified herein. The frame shall have sufficient strength and rigidity to enable the sander to perform at its maximum capacity without permanent deformation. The frame shall have access openings, with covers or hinged doors, to facilitate maintenance of all enclosed components requiring adjustment, lubrication and maintenance. Means shall be provided to securely fasten the sander to the floor or foundation.

3.4.2 Disk. The disk shall be of carbon steel plate with a surface for the attachment of an abrasive disk with adhesive. The disk shall be driven either by the motor shaft or by pulleys and V-belt drive. A flanged hub installed on the driven shaft shall facilitate disk installation and removal. Spindle speed shall not exceed 1250 RPM.

3.4.2.1 Disk table. The disk table shall be a one-piece iron alloy or semi-steel casting. The table top surface shall be flat within 0.005 inch.

3.4.2.2 Disk table adjustments. The table shall be mounted to permit tilting of the table as specified in table II. The vertical adjustment shall comply with the requirements specified in table I and shall be controlled by an adjustment screw and hand crank or hand wheel located in a position convenient to the operator. The transverse adjustment, when required, (see 6.2.1), shall be of the hand wheel and screw type, operable at the operation sides of the work table. If the veritcal adjustment range will allow the worktable to be lowered below the bottom edge of the disk to allow disk removal, the transverse adjustment may be omitted.

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3.4.2.3 Disk guard. A metal guard shall be mounted on the sander covering the back side and top edge of the outside periphery of the disk.

3.4.2.4 Dust chute. A dust chute shall encase the lower part of the sanding disk and shall be constructed to connect to an exhaust duct.

3.4.3 Spindle sander.

3.4.3.1 Spindle unit. The spindle unit shall consist of combination spindle drive and oscillating mechanism, a dynamically balanced spindle with spacing collars, and a hex nut or a capscrew with a washer, to install and allow removal of different size sanding drums with diameters conforming to table I.

3.4.3.1.1 Oscillating unit. The oscillating unit shall be constructed with a leakproof, internally lubricated drive mechanism . The oscillating unit shall be driven either directly by the motor or thru a gear train. The oscillating unit shall drive the spindle by means of an output shaft throughout the oscillating distance specified in table I.

3.4.3.1.2 Clutch. When a single motor is used to power both the disk and the spindle, mechanical means shall provide engagement and disengagement of power to the oscillating unit.

3.4.3.1.3 Spindle. The spindle shall be constructed of high carbon steel to accommodate spindle drums 9 inches long. The spindle shall be either threaded or shall have a self-holding taper conforming to ANSI B5.10.

3.4.3.1.4 Spindle drum. Not less than three different diameter spindle drums constructed of rubber shall be furnished with the sander. The three drums shall be of the diameter and length specified in table I and shall be constructed to uniformly support an abrasive sleeve, when the drum is tightened to the spindle.

3.4.3.1.5 Spindle table. The spindle table shall be one piece iron alloy or semi-steel casting with dimensions conforming to table 1. The table top surface and edges shall be flat to 0.005 inch.

3.4.3.1.6 Spindle clearance rings. A clearance ring for each diameter of spindle drum furnished shall interchangeably fit an opening in the spindle table to maintain spindle operating clearance.

3.4.4 Table tilt mechanism. The tilt mechanism shall provide a table tilting range of not less than the requirements of table II. A scale graduated in one degree increments, on the front of the sander, shall display the degrees of table tilt. Each five degrees along the scale shall be identified progressively by corresponding numerals. The tilting mechanism shall be manually controlled by a hand wheel or crank and shall include a securing device.

3.5. Characteristics and capacities. Unless otherwise specified (see 6.2), the characteristics and capacities of the sander shall be not less than those specified in table I.

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TABLE I

Sander Characteristics	
Characteristics	Capacities
Disk:	
Outside diameter, inches	30
Thickness, inches	5/8
RPM	1,050
Disk Table:	
Size, inches	15 x 38
Vertical adjustment, inches	8
Transverse adjustment, when required, inches	3
Spindle:	
Oscillating distance, inches	1
Strokes, per minute (oscillating)	50
Diameter of spindle drums, inches	2-3-4
Length of spindle drums, inches	9
RPM	1,725
Spindle Table:	
Size, inches	24 x 24

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TABLE II

Table Tilt Requirements		
Table	Table Tilt Setting	Resulting Sanded Angle (degrees)
Disk Table	0 degree (no tilt)	90± 1/2
	10 degrees toward the disk (up)	80± 1/2
	22 degrees away form the disk (down)	112± 1/2
	45 degrees away fromthe (down)	135± 1/2
Spindle Table	0 degrees (no tilt)	90± 1/2
	10 degrees toward the frame (up)	80± 1/2
	22 degrees away from the frame (up)	112± 1/2
	45 degrees away from the frame (down)	135± 1/2 135± 1/2

3.6 Electrical equipment.

3.6.1 Motors. Either a 5 hp motor shall drive both the disk and spindle or, a 5 hp motor shall drive the disk and a 3/4 hp motor shall separately drive the spindle. Motors shall be of the totally enclosed fan cooled type and shall have permanently lubricated bearings. Unless otherwise specified (see 6.2.1) motors shall be 230/460 volt, 3 phase, 60 hertz initally wired for operation on 230 volts. Motor characteristics shall conform to NEMA Standard Publication MG-1.

3.6.2 Controls. The sander shall be completely wired and ready for connection to the specified power source. Unless otherwise specified (see 6.2.1), the controls shall conform to NEMA Standard Publication IC-1 and shall be installed in a NEMA Type 12 enclosure. Unless otherwise specified (see 6.2.1), the motors shall have a magnetic starter with overload and under voltage protection. A transformer mounted on the sander shall provide voltage not to exceed 115 volts to the control circuit. A start-stop button shall be installed on the machine, convenient to the operator's position. Restarting shall require a deliberate action by the operator in the event of power failure.

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3.7 Equipment.

3.7.1 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of wrenches normally furnished by the manufacturer for the machine.

b. Six each fine, medium, and coarse abrasive disk, and six each fine, medium, and coarse abrasive sleeves for each size of spindle drum furnished with the sander.

c. One gallon can or one pound stick adhesive suitable for cementing the abrasive disk to the metal disk of the sander.

3.7.2 Optional equipment. Optional equipment shall be as specified (see 6.2.1.).

3.7.2.1 Duplicating attachment. When specified (see 6.2.1), the disk table shall be provided with a duplicating attachment which shall consist of at least an adjustable fence attached to an alignment bar sliding in a T-slot or groove, running the full depth of the worktable, perpendicular to the disk, and movable to and from the disk by a hand lever. The fence shall have a swing adjustment of at least 45 degrees on both sides of the centerline of the alignment bar with the adjustments of the swing marked in degrees on or about the base of the fence.

3.7.2.2 Circular segment sanding attachment. When specified (see 6.2.1), the disk table shall be provided with a circular segment sanding attachment. It shall consist of at least a metal plate having a minumum length equal to the width of the worktable, designed for sliding in T-slots or grooves machined in the disk table, parallel with and perpendicular to the disk running the full length and width of the disk table. The metal plate shall have a series of holes staggered throughout its entire length with a matching pivot pin for circular sanding accommodating diameters having radii from one inch through a dimension equal to the width of the disk table.

3.7.2.3 Chain hoist. When specified (see 6.2.1), a one ton capacity chain hoist, securely mounted to the machine, shall provide for swinging the disk clear of both work tables.

3.8 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate shall be securely affixed to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the nameplate shall contain the model number of the basic standard machine.

Nomenclature

Manufacturer's name

Manufacturer's model designation

Power input characteristics and ratings

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Date of manufacture

Contract Number of Order Number

National Stock Number or Plant Equipment Code

3.9 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to the sander. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application
Service interval
Type of lubricant

3.10 Workmanship. Workmanship of the machine and accessories shall be commensurate with the requirements specified herein and shall be of a quality prevailing among manufacturers normally producing equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First Article inspection. When First Article inspection is required, it shall be applied to the First Article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), First Article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the First Article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the

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contract. Quality conformance inspection shall consist of the examination in 4.5, the test in 4.6.1. Failure of the item to pass the examination or test shall be cause for rejection.

4.5. Examination. The machine and equipment shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification.

4.6. Tests.

4.6.1. Operational test. The sander shall be operated at no load for not less than 30 minutes. Operation of all controls, motors, adjusting mechanisms and other accessories shall be provided during the trial period.

4.6.2 Performance test. The instruments and methods used in performing the tests shall be at the option of the contractor provided they accurately determine the tolerances of the sander. Sander characteristics shall conform to tables I and II. Each sander shall be capable of performing the following tests.

4.6.1 Operational accuracy test. The sander shall sand a sample block of wood for each table tilt setting specified in table II. The sanded angle resulting from each table tilt setting shall conform to the degree of accuracy specified in table II.

4.6.2 Disk operating speed test. The operation speed of the disk shall be determined by means of a stroboscope or tachometer. Operation of the disk shall result in a stroboscope or tachometer reading equivalent to not less than 1,050 RPM and not more than 1,250 RPM.

4.6.2.3 Spindle operating speed test. The operating speed of the spindle shall be determined by means of a stroboscope or shall be calculated from the rated motor speed and, if applicable, the drive gear ratio. Stroboscopic determination or speed input calculation, as specified above, shall result in a spindle operating speed of not less than 1,750 RPM.

4.6.2.4 Spindle oscillation rate test. The number of spindle oscillations shall be tested for a least two 1-minute time periods, manually, with a stopwatch. Operation of the spindle shall result in a cycle count of not less than 50 oscillations per minute.

4.6.2.5 Spindle stroke length test. The spindle drive shaft shall be turned manually to operate the spindle through a complete stroke. The length of the stroke shall measure not less than 1 inch.

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4.6.2.6 Disk table vertical adjustment test. The disk table, without being tilted, shall be raised and lowered to the maximum limits. The vertically adjustable distance shall be not less than 8 inches. The upper limit shall be above the center of the disk and the lower limit shall be below the center of the disk.

4.6.2.7 Flatness test of table top.. Flatness of the table top shall be determined by placing a straightedge in at least two directions, at right angles to one another, on each table top, then checking for flatness with feeler gages. The table top shall be flat within 0.005 inch.

5. PACKAGING

5.1 Preservation, packing and marking. Unless otherwise specified (see 6.2.1), perservation, packing and marking shall conform to the requirements of MIL-STD-1188.

6. NOTES

6.1 Intended use. The machines covered by this specification are intended for use in workworking shops where repeated accuracies and tolerances are required.

6.2 Ordering data.

6.2.1 Acquisition documents. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First article, if required (see 3.1).

c. Specify additional safety and health requirements (see 3.2.3 and 6.3).

d. Painting, if different (see 3.3.4).

e. Characteristics and capacities, if different (see 3.5).

f. Voltage characteristics, if different (see 3.6.1).

g. Electrical controls, if different (see 3.6.2).

h. Type starter and voltage protection, if different (see 3.6.2).

i. Standard equipment, if required (see 3.7.1).

j. Optional equipment, if required (see 3.7.2).

k. Duplicating attachment, if required (see 3.7.2.1).

l. Circular segment sanding attachment, if required (see 3.7.2.2).

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m. Chain hoist, if required (see 3.7.2.3).

n. Nameplate, if different (see 3.8).

o. Lubrication charts or plates, if different (see 3.9).

p. First article inspection, if different (see 4.3).

q. Preservation, packing and marking, if different (see 5.1).

6.2.2 Contract data requirements. Required technical data, such as operators manuals, parts lists, wiring diagrams, and other instruction for operation and maintenance as identified on a numbered DD Form 1664 should be specified on a DD Form 1423 incorporated into the contract.

6.3 Safety and health requirements. Paragraph 3.2.3 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total level of these hazards into the environment (and does not limit the hazard level of individual machines in the environment) the requesting activity is advised to analyze the existing hazard levels in the proposed operating environment. If specific point-of-operation guarding is required, as in most cases, the guard configuration is dependent on the size and configuration of workpieces. The above, and any other additional safety and health requirements, should be specified in detail under 6.2.1 (c).

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - SH
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Preparing Activity:
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Project Number:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-S-26753C, Sander, Disk & Spindle, Woodworking

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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